

Name: _____

Please **CIRCLE** the tutorial session you attend.

Mon. 1.30-2.20; Mon. 2.30-3.20; Mon. 3.30-4.20; Mon. 4.30-5.20

Tue. 9.30-10.20; Tue. 10.30-11.30; Tue. 1.30-2.20

Fri. 1.30-2.20; Fri. 2.30-3.20

Each question is worth 1 mark.

1. Which of the following correctly identifies a difference between cross-sectional data and time series data?
 - a. Cross-sectional data is based on temporal ordering, whereas time series data is not.
 - b. Time series data is based on temporal ordering, whereas cross-sectional data is not.
 - c. Cross-sectional data consists of only qualitative variables, whereas time series data consists of only quantitative variables.
 - d. Time series data consists of only qualitative variables, whereas cross-sectional data does not include qualitative variables.

2. Which of the following is an assumption on which time series regression is based?
 - a. A time series process follows a model that is nonlinear in parameters.
 - b. In a time series process, no explanatory variable is a perfect linear combination of the others.
 - c. In a time series process, at least one independent variable is a constant.
 - d. For each time period, the expected value of the error u_t , given the explanatory variables for all time periods, is positive.

3. The model: $y_t = \beta_0 + \beta_1 x_{t1} + u_t$, where $t = 1, 2, \dots, n$ is an example of a(n):
 - a. autoregressive of order one model.
 - b. static model.
 - c. finite distributed lag model.
 - d. infinite distributed lag model.

4. Refer to the following model:
 $y_t = \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} + \delta_3 x_{t-3} + u_t$, where $t = 1, 2, \dots, n$.
This is an example of a(n):
 - a. infinite distributed lag model.
 - b. finite distributed lag model of order 1.
 - c. finite distributed lag model of order 2.
 - d. finite distributed lag model of order 3.

5. In the presence of serial correlation in error terms:
 - a. estimated standard errors remain valid.
 - b. estimated test statistics remain valid.
 - c. estimated OLS values are not BLUE.
 - d. estimated variance does not differ from the case of no serial correlation.

6. Which of the following statements is true?
 - a. When serial correlation exists in the u terms, the usual t test is valid.
 - b. When explanatory variables include a lagged dependent variable, the Durbin Watson test for testing serial correlation is valid.
 - c. Breusch-Godfrey test can be used to check for second order serial correlation, i.e., $Cov(u_t, u_{t-2})$.
 - d. Breusch-Pagan test can be used to check for first order serial correlation, i.e., $Cov(u_t, u_{t-1})$.

7. In the presence of heteroskedasticity, the usual OLS estimates of:
- standard errors are valid, whereas the t statistics and F statistics are invalid.
 - t statistics are valid, but the standard errors and F statistics are invalid.
 - F statistics are valid, but the standard errors and t statistics are invalid.
 - standard errors, t statistics, and F statistics are invalid.
8. The error term in a simple (cross-sectional) regression model is said to exhibit heteroskedasticity if _____.
- $Var(u|x_1)$ is not constant
 - $Var(\hat{\beta}_1|x_1)$ is not constant
 - $Var(y|x_1)$ is not constant
 - (a) and (c)
9. Consider an estimated regression model $\hat{y} = \hat{\beta}_0$, where $\hat{\beta}_0 = \bar{y} = 25$. If the total sum of squares (SST) is 100, calculate the R^2 value.
- 0.00
 - 0.25
 - 0.75
 - 1.00
10. Consider the following cross-sectional regression model: $y_i = \beta_0 + \beta_1 x_i + u_i$, where $i = 1, 2, \dots, n$. Find out the correct statement(s).
- The sum of the OLS residuals could be negative, positive or zero
 - The sample covariance between the explanatory variable x and the OLS residuals is zero.
 - The point $(\bar{x}, \bar{y})$ always lies on the OLS regression line.
 - (b) and (c)

True/False Questions

- ____ 11. The homoskedasticity assumption in time series regression implies that the variance of the error term cannot be a function of time.
- ____ 12. The serial correlation–robust standard errors are typically larger than the usual OLS standard errors when there is serial correlation
- ____ 13. In time series regressions, it is advisable to check for serial correlation first, before checking for heteroskedasticity
- ____ 14. Consider a regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. If the Gauss Markov assumptions hold, then $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2$ is an unbiased estimator of σ^2 .
- ____ 15. Suppose X and Y are two correlated random variables. Then, $Var(3X - 2Y) = 9Var(X) + 4Var(Y) - 2Cov(X, Y)$.